Lecture 20 // Oct. 29 // The Casson Invariant

Last time, we revisited the character variety.
Today, we'll discuss the Casson inv. For a $\mathbb{Z}HS^3$ Y, the Casson inv. $\lambda(Y)$ is, up to a constant factor, "the # of irred $SU(2)$ rep's of $\pi_1(Y)$". It is related to IFH_* by a famous thm, due to Taubes:

Thm [Taubes]

$$\frac{1}{2}\chi(H_*(C(Y))) = \lambda(Y)$$

where $\chi(H_*(C(Y)))$ is the Euler char. of the irred. chain cpx

You might ask: isn't this tautological? Given a free, fin-gen cpx, we know that the EC of the cpx is the EC of the H_*, and both objects were literally defined as a count of $SU(2)$ rep^{n}s.

However, the point is that λ is acc. a completely diff way to count $SU(2)$ rep^{n}s which is very classical and existed long before IFH_*.

We won't prove this thm, but we will discuss its def^n and a short hist. overview.

What are some interesting properties of λ?

1) It sat's a formula. Doing $1/n$ surg. on some $K \subset \mathbb{Z}HS^3$ gives another $\mathbb{Z}HS^3$, and the CI's are rel'd by

$$\lambda(Y_{\frac{1}{n+1}}(K)) - \lambda(Y_{\frac{1}{n}}(K)) = \frac{1}{2}\Delta_K''(1)$$

where Δ_K'' is the 2nd derivative of the Alex poly eval'ed at 1.

Of course, when $n=0$, $Y_{\frac{1}{0}} \cong Y$, and it turns out that every $\mathbb{Z}/\text{HS}^3$ can be obtained from S^3 by a seq of ± 1 surg on knots.

This tells us that CI is purely combinatorial (eg. there's no gauge theory involved!), relying only on the Alex poly of knot.

2) CI is a $\mathbb{Z}$-lift of the Rokhlin Invt:

$$\lambda(Y) \equiv \mu(Y) \bmod 2$$

(In fact, this is implied by (1), but we won't prove this.)

Historically, interest in the CY was related to using the RI as a poss. disp't. of the 3DPC.

Before the advent of floer/gauge theory, the hope was finding a htpy S^3 with nontriv. RI. But the CI says no: if you believe that the CI is a count of rep's, any htpy S^3 has triv $SU(2)$ rep's, so the RI is 0.

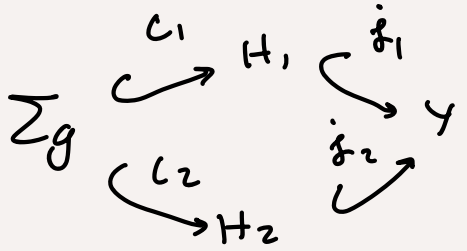
Rmk Every $Y = \pm$ surg on link of unknots, but Δ_K is computed at each step, so $\Delta_K \neq 1$.

So what is this mysterious invt? We said that the idea was to count conj. classes of $SU(2)$ rep's. How do we do this?

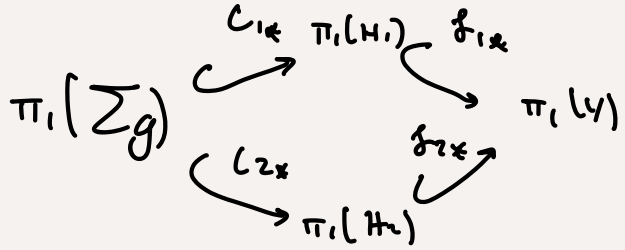
Ofc, the problem that we've seen is that the Char-V $R(Y)$ may not be discrete – what does "counting" mean here?

What we mean by "count" is an intersection #.

The idea is very geometric. Spse Y is a 3MH S^3. Choose a Heeg split of $Y = H_1 \cup_{\Sigma_g} H_2$. We have the inclusions



which induce maps on the fund gps

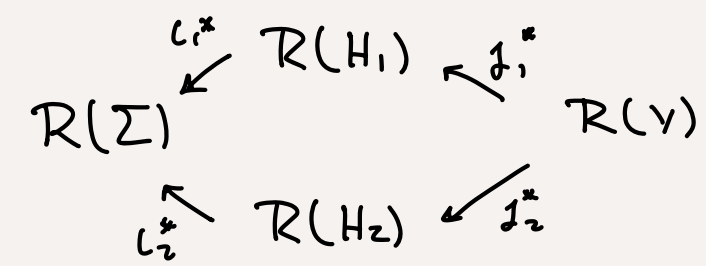


which are clearly surjective.

Now, we introduce the following notation

Def
- The Representation Variety: $R(Y) := \mathrm{Hom}(\pi_1(Y), SU(2))$
- The Irred. Representation Variety: $R^{irr}(Y) =$ Homs w/ stab $\pm I$
- The Character Variety: $\mathcal{R}(Y) = R(Y)/\mathrm{conj}$
- The Irred. Character Variety: $\mathcal{R}^{irr}(Y) = R^{irr}(Y)/\mathrm{conj}$

Applying $\mathrm{Hom}(-, SU(2))$ to every group in the diagram reverses the arrows, giving injective maps



This process preserves subgroups and is conjugation equivariant, so we could replace the "R"'s in the diagram with any choice of $\{R^{irr}, R, R^{irr}\}$

What the diagram is spoked to suggest is that $R^{irr}(Y)$ can be identified w/ the intersection of the H_i's in Σ:

$$R^{irr}(Y) \cong (i_1^*) R^{irr}(H_1) \cap (i_2^*) R^{irr}(H_2) \text{ inside } R^{irr}(\Sigma)$$

Since every rep^n on $\pi_1(Y)$ must split into rep^{n}s on $\pi_1(H_1)$ and $\pi_1(H_2)$ which agree over their intersection. This is good b/c these are very simple!

Prop $R^{irr}(H_1)$ is an open mfd of dim $3g-3$

Pf First note that since $\pi_1(H_1) \cong *_g \mathbb{Z}$, $R(H_1) = SU(2)^g$ though $R^{irr} \subset R$ is complicated. Therefore

$$\begin{aligned} \dim(R^{irr}(H_1)) &= \dim(SU(2)^g) - \dim(\text{Conj action}) \\ &= 3g - \dim(SU(2)) \\ &= 3g-3. \end{aligned}$$

Being red. is a closed condition, so R^{irr} is open. □

Prop $R^{irr}(\Sigma)$ is an open mfd of dim $6g-6$

pf Recall that $\pi_1(\Sigma_g) = \langle a_1, b_1, \dots, a_g, b_g \mid \prod [a_i, b_i] = 1 \rangle$, which we can interpret as $f^{-1}(1)$ for the map

$$f: \underset{2g}{*}\mathbb{Z} \longrightarrow \mathbb{Z}$$
$$(a_1, b_1, \dots, a_g, b_g) \longmapsto \prod [a_i, b_i]$$

For any rep^n, this gives a map

$$f_\rho : SU(2)^{2g} \to SU(2)$$
$$(A_1, B_1, \dots, A_g, B_g) \longmapsto \prod [A_i, B_i]$$

so $\rho(\pi_1(\Sigma)) = f_\rho^{-1}(I)$. It turns out that Df is surj for irred rep^{n}s, so $\dim(R^{irred}(Y)) = 6g-3 - \dim(SU(2))$
$= 6g-6$

Similarly being reducible is a closed condition, so $R^{irr}(Y)$ is open. □

Def Choose a compactly supp. perturbation (or isotopy) of $R^{irr}(H_a)$ inside $R(\Sigma)$ such that

$$(L_1^*)\, R^{irr}(H_1) \pitchfork (L_2^*)\, R^{irr}(H_2)$$

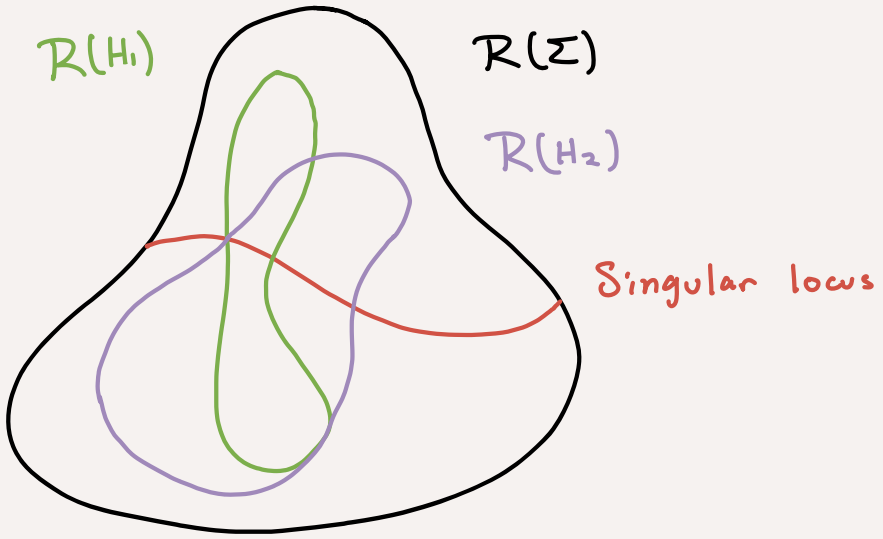
The Casson Inv. is

$$\lambda(Y) = \left(\frac{(-1)^g}{2} \right) \cdot \left(\begin{array}{c} \text{Signed \# of intersections btwn} \\ (L_1^*)\, R^{irr}(H_1) \text{ and the perturbed} \\ (L_2^*)\, R^{irr}(H_2) \end{array} \right)$$

This definition begs a few questions!

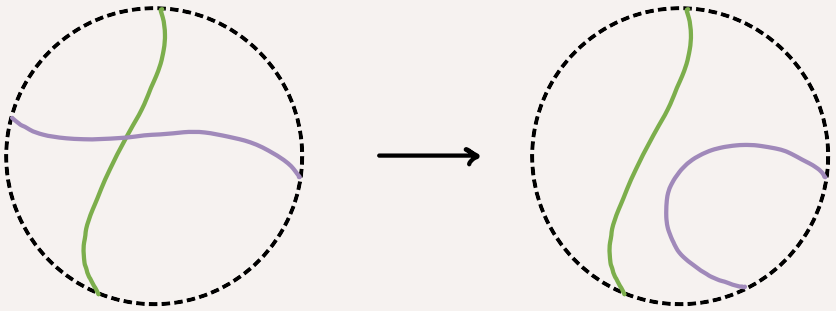
It turns out that the rep^n vars can be oriented, so we can take a signed count of intersections. The factor of $(-1)^g$ is to account for different choices of Heeg splittings, though why the signed count is divisible by 2 is not obvious..

However you should ask: What's up w/ needing a cpt supp perturbation? The point here is that $R(\Sigma)$ is not a smooth manifold! It contains a singular locus of reducible rep^{n}s, which we must avoid when performing our signed count since the ambient space is not a manifold. This introduces some complexities.



We need to perturb in the first place to ensure transversality. If these perturbations can be supported everywhere, who's to say we can't perturb every intersection away! Also, we must take our perturbation to be away from the singular locus. In other words, if the perturb isn't compactly supported, it need not fix the endpoints of $R(H_2) \cap$ open ball.

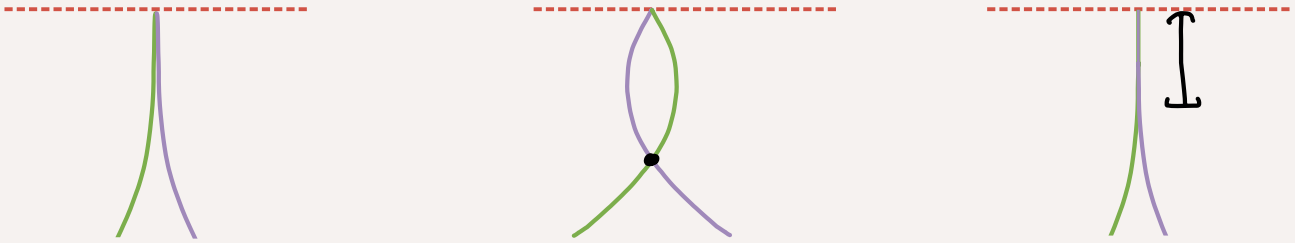
The local picture is:



Intersection # is 1 Intersection # is 0

The one caveat is that restricting the type of perturbation restricts the type of intersections we can perturb away! That is, if the intersection set itself is not compact, we're stuck. This means that the set is open - so they intersect in the singular locus as well.

Given that $R(H_1)$ and $R(H_2)$ intersect at a reducible rep^n, lets consider what they could look like nearby:



Int # is 0 Int # is 1 Intersection is a half-open interval & non-transverse, can't perturb away

both o.k.

Claim The third case can't happen

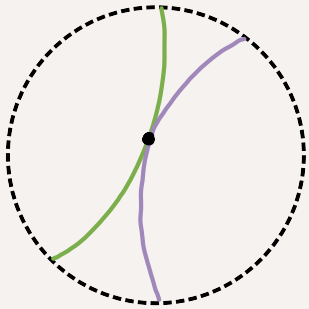
The point here is that we need to prove some analytic properties to make sense of the Int #, since we're not working in a smooth manifold.

Prop For a $\mathbb{Z}HS^3$, $R(H_1)$ and $R(H_2)$ intersect in the sing locus in exactly one pt

Pf For the trivial representations of $\pi_1(H_1)$ and $\pi_1(H_2)$. Clearly they both restrict to the triv. rep^n of $\pi_1(\Sigma)$, so they are an element of $R(Y)$. Since the id is stabilized by the entire group, it is in the singular locus.
Any other rep^n in this locus must have nontriv stab, so it must be abelian. Thus the rep^n factors through $\pi_1^{ab}(Y) = H_1(Y) = 0$ since Y is a $\mathbb{Z}HS^3$. □

that for the $n=3$ Brieskorn spheres,

However, recall from last time that we claimed the set of pts was an honest conn without perturbation. This means that $R(H_1)$ and $R(H_2)$ can't intersect non transversely, even if the $\wedge$ is discrete:



Prop For the $n=3$ Brieskorn sphere, the "naive" pt count is the correct one for λ

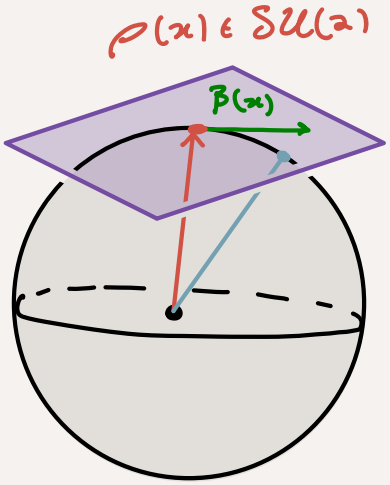
But how can we even verify this? Or, what even are the tangent spaces, explicitly? How do we calculate their intersection? In order to discuss $\wedge$'ity, it will be convenient to introduce some language from Alg Geo.

Interlude:

Let π be a fin pres group, thought of as $\pi = \pi_1(X)$.
Let $\rho: \pi \to SU(2)$ be a rep^n.

What is $T_\rho R(\pi)$, ie, the tan. space to the rep^n variety at the rep^n ρ? If we forget, for a second, that ρ is a gp hom, ρ is just some map from π to $SU(2)$. Given $\rho \in \mathrm{Map}(\pi, SU(2))$, we can ask the more general question: what is $T_\rho \mathrm{Map}(\pi, SU(2))$?

A tan space at a point can be thought of as the space of dirs you can move (perturb) in to stay in whatever locus the point is in. Explicitly, for some $x \in \pi_1$, a "nearby" map to $\rho(x)$ is of the $\rho(x) + \varepsilon\beta(x)$, for $\beta(x) \in T_{\rho(x)} SU(2)$



But its more convenient to remember that right mult moves $T_1 SU(2)$ to $T_{\rho(x)} SU(2)$, so we can think of $\beta(x) \in T_{\rho(x)} SU(2)$, instead as $\beta(x) = \eta(x)\rho(x)$, for $\eta(x) \in T_1 SU(2) = \mathfrak{su}(2)$

Therefore the new point is

$$\rho(x) + \varepsilon\beta(x) = \rho(x) + \varepsilon\eta(x)\rho(x)$$
$$= (1 + \varepsilon\eta(x))\rho(x).$$

Now we have an explicit understanding of what it means to be in $T_\rho \mathrm{Map}(\pi, SU(2))$. But not every map is a rep^n. So we must ask which maps are homomorphisms, at least to 1st order. Lets linearize the condition $\rho(xy) = \rho(x)\rho(y)$.

For a perturb. $\eta(x): \Pi \to \mathfrak{su}(2)$, we need

$\rho(xy) + \varepsilon M(xy)\rho(xy)$ to equal (up to first order)

$$\left(\rho(x) + \rho(x)\varepsilon\eta(x)\right)\left(\rho(y) + \rho(y)\varepsilon\eta(y)\right).$$

Expanding, $\left(\rho(x) + \rho(x)\varepsilon\eta(x)\right)\left(\rho(y) + \rho(y)\varepsilon\eta(y)\right)$

$$= \rho(x)\rho(y) + \rho(x)\varepsilon\eta(x)\rho(y) + \rho(x)\rho(y)\varepsilon\eta(y) + \underbrace{\rho(x)\varepsilon\eta(x)\rho(y)\varepsilon\eta(y)}_{2^{nd}\text{ order}}$$

So if $\rho(xy) + \varepsilon M(xy)\rho(xy)$

$$= \rho(x)\rho(y) + \rho(x)\varepsilon\eta(x)\rho(y) + \rho(x)\rho(y)\varepsilon\eta(y)$$

and if we know $\rho(xy) = \rho(x)\rho(y)$, $\rho(x)\eta(x) = \eta(x)\rho(x)$, $\rho(y)\eta(y) = \eta(y)\rho(y)$,

Then

$$\varepsilon M(xy)\rho(xy) = \rho(x)\varepsilon\eta(x)\rho(y) + \rho(x)\rho(y)\varepsilon\eta(y)$$

$$\eta(xy)\rho(x)\rho(y) = \rho(x)\eta(x)\rho(y) + \rho(x)\rho(y)\eta(y)$$

$$\eta(xy)\rho(x)\rho(y) = \rho(x)\eta(x)\rho(y) + \rho(x)\rho(y)\eta(y)$$

$$\eta(xy) = \rho(x)\eta(x)\rho(y)\rho(y)^{-1}\rho(x)^{-1} + \rho(x)\rho(y)\eta(y)\rho(y)^{-1}\rho(x)^{-1}$$

$$\eta(xy) = \eta(x) + \rho(x)\eta(y)\rho(x)^{-1}$$

Setting $\mathrm{Ad}_{\rho(x)}\eta(y) := \rho(x)\eta(y)\rho(x)^{-1}$, we can conclude that

$$\eta(xy) = \eta(x) + \mathrm{Ad}_{\rho(x)}\eta(y).$$

This gives an "algebraists" explicit description of the tangent space of a rep^n in rep^n space.

Def The Zariski tangent space

$$Z^1_\rho = \left\{ \eta : \pi \to su(2) \;\middle|\; \eta(xy) = \eta(x) + Ad_{\rho(x)}\eta(y) \right\}$$

This is a generalized notion of tangent space (which is) defined at sing points, but if ρ is a sm pt of $R(\pi)$, then Z^1_ρ is the tan space, so this recovers the usual notion of the tangent space.

Rmk Z doesn't stand for Zariski, and the 1 doesn't mean 1st order... the reason for this notation will soon become clear.

Now, we wish for a def^n of the tan space in the Char Var. We get $\mathcal{R}$ from R by quotienting by the $SU(2)$ action, so lets try to quotient Z^1_ρ by this action as well.

What is $T_\rho\left(\begin{array}{c}\text{orbit of } \rho \text{ under} \\ \text{conj action}\end{array}\right)$?

Fix $u \in su(2)$. Conjugating ρ by an element which is v. close to the id sends ρ to $(1+\varepsilon u)\rho(1-\varepsilon u)$. Expanding,



$$(1+\varepsilon u)\rho(1-\varepsilon u) = \rho + \varepsilon u\rho - \rho u.$$
$$= 1 + \varepsilon(u - Ad_\rho u)\rho$$

So when we act by a small gauge trans, the direction ρ moves along in is $Ad_\rho u = \rho(x) u \rho(x)^{-1}$

Def $B^1_\rho = \left\{ \eta : \pi \to su(2) \mid \exists \ u \in su(2) \text{ s.t. } \eta(x) = u - A d_{\rho(x)} u \ \forall x \right\}$

Putting this all together, we have discovered that

$$T_{[\rho]} \mathcal{R}(\pi) = Z^1_\rho / B^1_\rho$$

Def The 1st group cohomology of π is

$$T_{[\rho]} \mathcal{R}(\pi) = Z^1_\rho / B^1_\rho$$

Explicitly, what we mean is the 1st gp coh, w/ elements in $su(2)$, twisted by ρ.

Rmk So Z & B are cocycles and coboundaries and 1 is 1st!

By construction,

$\dim\left(T_{[\rho]} \mathcal{R}(\pi) \right)$ is the exp. dim of the rep^n space at that rep^n.

Back to earth: how does this relate to the char var.?

Lemma Let ρ be an irred rep^n coming from $\mathcal{R}^{im}(H_1) \cap \mathcal{R}^{irr}(H_2)$, where $\mathcal{R}^{irr}(H_2)$ is not perturbed.

This $\cap$ is $\pitchfork$ at ρ IFF $H^1_\rho(\pi, su(2)) = 0$

Pf We do some slight abstract nonsense:

There is a LES for gp coh:

$$H^0_\rho(\pi_1(\Sigma), \mathfrak{su}(2)) \to H^1_\rho(\pi_1(Y), \mathfrak{su}(2)) \to$$

$$H^1_\rho(\pi_1(H_1), \mathfrak{su}(2)) \oplus H^1_\rho(\pi_1(H_2), \mathfrak{su}(2)) \xrightarrow{C_1^* \oplus C_2^*} H^1_\rho(\pi_1(\Sigma), \mathfrak{su}(2))$$

The $\pitchfork$ cond. is precisely that the map $C_1^* \oplus C_2^*$ is surjective. At a smooth point, we asserted that this constr. was the same as the normal bdl space, so $\dim\left(H^1_\rho(\pi_1(H_i), \mathfrak{su}(2))\right) = 3g-3$ and $\dim\left(H^1_\rho(\pi_1(\Sigma), \mathfrak{su}(2))\right) = 6g-6$.

Therefore $C_1^* \oplus C_2^*$ is surjective iff it is an iso.

We haven't yet defined the 0th gp coh:

$$H^0_\rho(\pi_1(\Sigma), \mathfrak{su}(2)) = \left\{ u \in \mathfrak{su}(2) \text{ s.t. } u - \mathrm{Ad}_{\rho(x)} u = 0 \ \forall x \right\}, \text{ which}$$

is nonzero if there exists an infinitessimal conj. which doesn't move the rep^n. If there existed such a u, it would have nontriv stab — but we're only working with irreds, so

$$H^0_\rho(\pi_1(\Sigma), \mathfrak{su}(2)) = 0.$$

Therefore $C_1^* \oplus C_2^*$ is an iso. $\square$

The moral here is to know whether we must perturb, it is enough to check if $H^0_\rho(\pi_1(\Sigma), \mathfrak{su}(2)) = 0$.